Step 0: Initial Document Check

- **0.1:** The document is an RCC structural drawing of "FOUNDATIONS" only.
- **0.2:** Site Location: Malpe Fishery Harbour, Udupi.
- **0.3:** All compliance checks are based only on IS 456:2000 and SP 34.

Step 1: Locate the "NOTES" Section

- **Status:** Compliant

Step 2 & 3: Extract and Verify Design Parameters from "NOTES"

Criteria	Extracted Value	Compliance Check	Status
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1. Grade of Concrete	M25	M25 is a suitable grade for general RCC works. Environmental exposure conditions for Malpe Fishery Harbour (coastal area) would typically require M30 or higher for severe/very severe conditions as per IS 456:2000 Table 5. M25 might be acceptable for moderate exposure. Without specific environmental exposure classification, it's difficult to definitively confirm compliance.	Cannot Verify
2. Reinforcement Bars	HYSD TMT Bars, Grade Fe 550, conforming to IS 1786-1985. Column vertical reinforcement: 8Y16. Column ties: Y8@200. Footing bottom reinforcement: Y10@125.	Fe 550 is a high-strength grade, which is good. IS 1786-1985 is an older standard; current standard is IS 1786:2008. Shear reinforcement (ties) are specified.	Non-Compliant
3. Lap Length	50 times dia of the bar, not more than 50% of bars are lapped at a section.	50d is a common and generally compliant value. The staggering requirement is also good practice.	Compliant
4. Clear Cover	Footing/Wall: 50 mm, Columns: 40 mm, Slab: 25 mm, Beam: 30 mm.	Footing cover of 50mm is the minimum as per IS 456:2000. The drawing shows 150mm PCC below the footing, which is good practice for a 50mm cover. Column cover of 40mm is compliant (40mm to 70mm). Slab and beam covers are also generally compliant for normal exposure.	Compliant
5. Development Length (Ld)	50 times the dia. of the bar.	50d is a common and generally compliant value.	Compliant
6. Safe Bearing Capacity (SBC) of soil	10 T/m2 (100 kN/m2). PCC is mentioned (150mm thick).	SBC is mentioned. PCC is specified.	Compliant
7. Seismic Zone and Wind Load	Seismic Zone III and Wind Forces as per IS 1893:2016 & IS 875:2015 respectively.	Seismic zone and wind load standards are mentioned and are current.	Compliant

8. Building Limitations	Structure is designed for Ground + 1 Storey + Lightweight Roof.	Building limitations are specified.	Compliant
9. Structure's Purpose	Entrance Gate and Guard Room.	The purpose of the structure is mentioned.	Compliant
10. Floor Heights	Not explicitly mentioned in the "NOTES" section. "First Floor to Terrace Lvl" is indicated in the column detail, but specific floor heights are not listed.	Specific floor heights are not provided.	Missing Information
11. Schedule of Footings	Present and consistent with F1 footing details.	A "SCHEDULE OF FOOTINGS" table is present and provides dimensions and reinforcement for F1.	Compliant
12. Footing Type	Isolated footings (F1).	Isolated footings are used for a Ground + 1 Storey structure, which is appropriate.	Compliant
13. Reinforcement in High-Rise Buildings	Not applicable.	The structure is Ground + 1 Storey, not a high-rise building.	Not Applicable
14. Raft Foundation Reinforcement	Not applicable.	Raft foundation is not used.	Not Applicable
15. Lift Design	Not applicable.	No lift is designed for this structure.	Not Applicable
16. Soil Improvement	"Improved S.B.C of soil considered is 10 T/m2". Note 1 in Section X mentions "SIZE STONE LAYERS BELOW THE FOOTING IS TO BE LAID AFTER EXCAVATING AND OMPACTING THE SAME AS PER THE STANDARD PROCEDURE".	Soil improvement is indicated by the improved SBC and the mention of stone layers and compaction.	Compliant
17. Column Ties	Column ties are shown in Section X and the C1 detail. Ties are Y8@200. They appear continuous in the C1 detail.	Column ties are shown and appear continuous.	Compliant
18. Plan of Ties	A separate plan for ties is not explicitly present, but the C1 column detail shows the arrangement of ties.	A dedicated plan for ties is not present, but the detail provides the necessary information.	Missing Information

19. Outer Ties Check	Column vertical reinforcement: 8Y16. Column ties: Y8@200. Footing bottom reinforcement: Y10@125. Percentage of steel in columns: For a 300x450 column, 8Y16 bars (Area = 8 *pi* (16/2)^2 = 1608.5 mm2). Gross area = 300 *450 = 135000 mm2. Percentage = (1608.5 / 135000)* 100 = 1.19%. This is >= 0.8%. Footing reinforcement: Y10@125. For a 2100mm wide footing, area of steel per meter = (1000/125) *(pi* (10/2)^2) = 8 *78.54 = 628.32 mm2. Percentage = (628.32 / (2100* 450)) * 100 = 0.066%. This is less than 0.12% for slabs/footings as per IS 456:2000 (cl. 26.5.2.1).	Column steel percentage is compliant. Footing steel percentage is non-compliant.	Non-Compliant
20. Cross-Section Area	Not explicitly stated whether gross or effective area is used for calculations in the notes.	This information is not explicitly mentioned in the notes.	Missing Information
21. Steel Curtailment	Not explicitly mentioned in the notes.	For a G+1 structure, curtailment might not be as critical as in high-rise buildings, but general principles of curtailment should be followed. Not explicitly detailed.	Missing Information
22. Maximum Steel Percentage in Columns	Not explicitly mentioned in the notes. Calculated percentage for C1 is 1.19%, which is well within the 6% (or 4% with lapping) limit.	The maximum percentage is not stated in the notes, but the calculated value is compliant.	Missing Information

Step 5: Report Missing or Wrong Information

Grade of Concrete: Cannot verify compliance without specific environmental exposure classification for Malpe Fishery Harbour. M25 might be insufficient for severe/very severe coastal conditions.

Reinforcement Bars: The standard for reinforcement bars (IS 1786-1985) is outdated; the current standard is IS 1786:2008.

Floor Heights: Specific floor heights are not explicitly mentioned in the notes.

Plan of Ties: A separate plan for ties is not present, although the column detail provides some information.

Outer Ties Check (Footing Reinforcement): The calculated percentage of steel for footing (0.066%) is less than the minimum required 0.12% for slabs/footings as per IS 456:2000 (cl. 26.5.2.1).

Cross-Section Area: It is not explicitly stated whether gross or effective cross-section area is used for calculations.

Steel Curtailment: Details regarding steel curtailment are not provided.

Maximum Steel Percentage in Columns: The maximum steel percentage allowed in columns is not explicitly stated in the notes.

Summary of Compliance

• **Total Criteria Evaluated:** 22

• **Compliant Items:** 10

• **Non-Compliant Items:** 2

• **Missing Information Items:** 6

• **Cannot Verify Items:** 1

• **Not Applicable Items:** 3

• **Overall Verdict:** The drawing has several missing pieces of information and one non-compliance regarding footing reinforcement, which is a critical structural element. The outdated IS code for reinforcement bars is also a concern. While the drawing is for foundations, the number of missing/non-compliant items (8 out of 22 evaluated, excluding N/A) indicates significant areas for improvement to ensure full compliance with IS 456:2000 and SP 34.